

CSE344 FINAL REPORT

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1. Introduction and Problem Definition

Objective

The goal of this project was to develop a multi-client chat server with support for:

- ◆ Real-time messaging (broadcast, private whispers)
- ◆ File transfers (with size and extension restrictions)
- ◆ Room-based communication (join/leave rooms)
- ◆ Concurrent client handling (using threads)
- ◆ Graceful disconnection handling

Challenges

- Thread synchronization (avoiding race conditions)
- Resource management (file descriptors, memory leaks)
- Unexpected disconnections (network failures, crashes)
- Scalability (handling multiple clients efficiently)

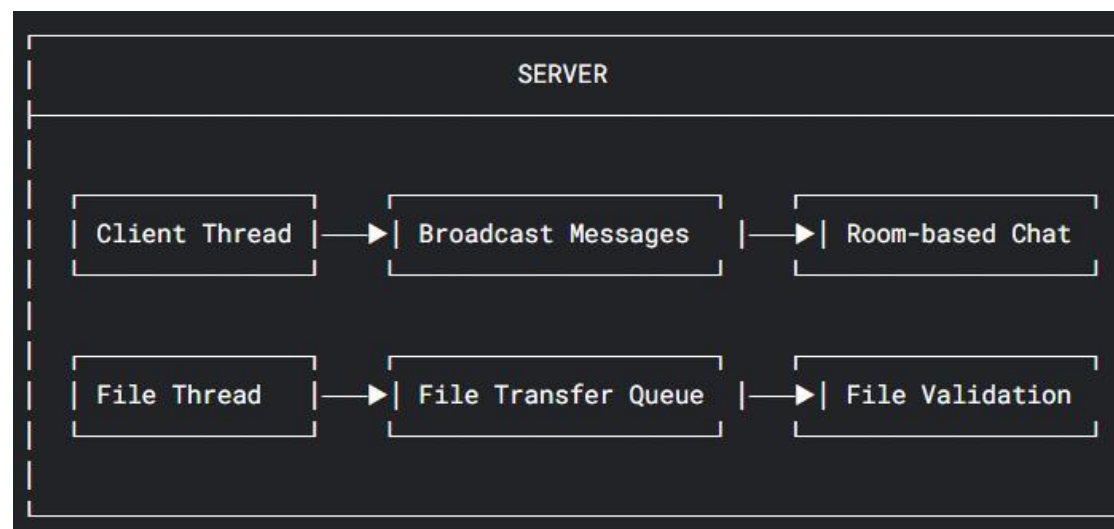
2. Design Details

- ✓ IPC (Inter-Process Communication)
- ✓ TCP Sockets for client-server communication.
- ✓ Thread-safe queues for file transfer requests.
- ✓ Mutexes & Semaphores for shared resource protection.

Thread Model

- ✧ **Main Thread** : Accepts new client connections
- ✧ **Client Handler - Threads (per client)** : Processes client commands (/join, /whisper, /sendfile)
- ✧ **File Handler Thread** : Manages file upload queue (MAX_UPLOAD_QUEUE = 5)

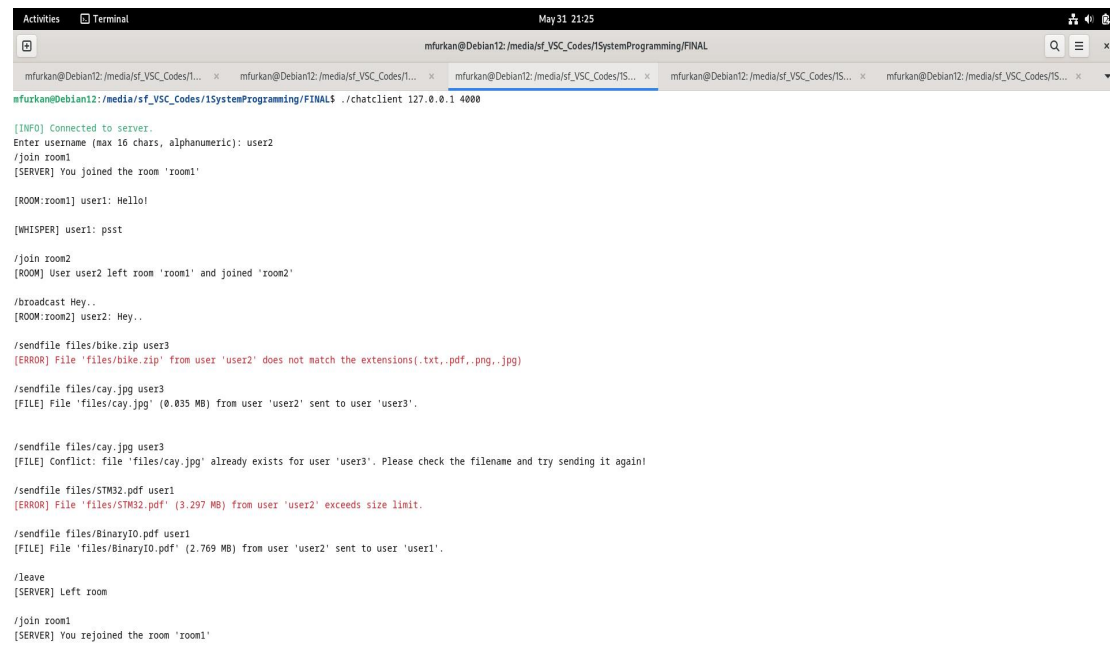
Architecture Diagram



3. Issues Faced and Solutions

● Issue	Solution
● Race conditions in client list access protect clients[] array	Used <code>pthread_mutex_t</code> to
● Memory leaks in file size strings error paths	Ensured <code>free(file_size)</code> in all
● File transfer conflicts	Implemented alert for same filenames
● Deadlocks in queue handling queue synchronization	Used <code>sem_t</code> for bounded

4. Test Cases and Results



A terminal window titled 'May 31 21:25' showing the output of a chat client. The user is in a room named 'room1' and has sent several messages. The chat client is connected to a server at 127.0.0.1:4000. The output shows the user's messages and the server's responses, including file upload errors and room management actions.

```
mfurkan@Debian12:/media/sf_VSC_Codes/ISystemProgramming/FINAL$ ./chatclient 127.0.0.1 4000

[INFO] Connected to server.
Enter username (max 16 chars, alphanumeric): user2
/join room1
[SERVER] You joined the room 'room1'

[ROOM:room1] user1: Hello!

[WHISPER] user1: psst

/join room2
[ROOM] User user2 left room 'room1' and joined 'room2'

/broadcast Hey..
[ROOM:room2] user2: Hey..

/sendfile files/bike.zip user3
[ERROR] File 'files/bike.zip' from user 'user2' does not match the extensions(.txt,.pdf,.png,.jpg)

/sendfile files/cay.jpg user3
[FILE] File 'files/cay.jpg' (0.035 MB) from user 'user2' sent to user 'user3'.

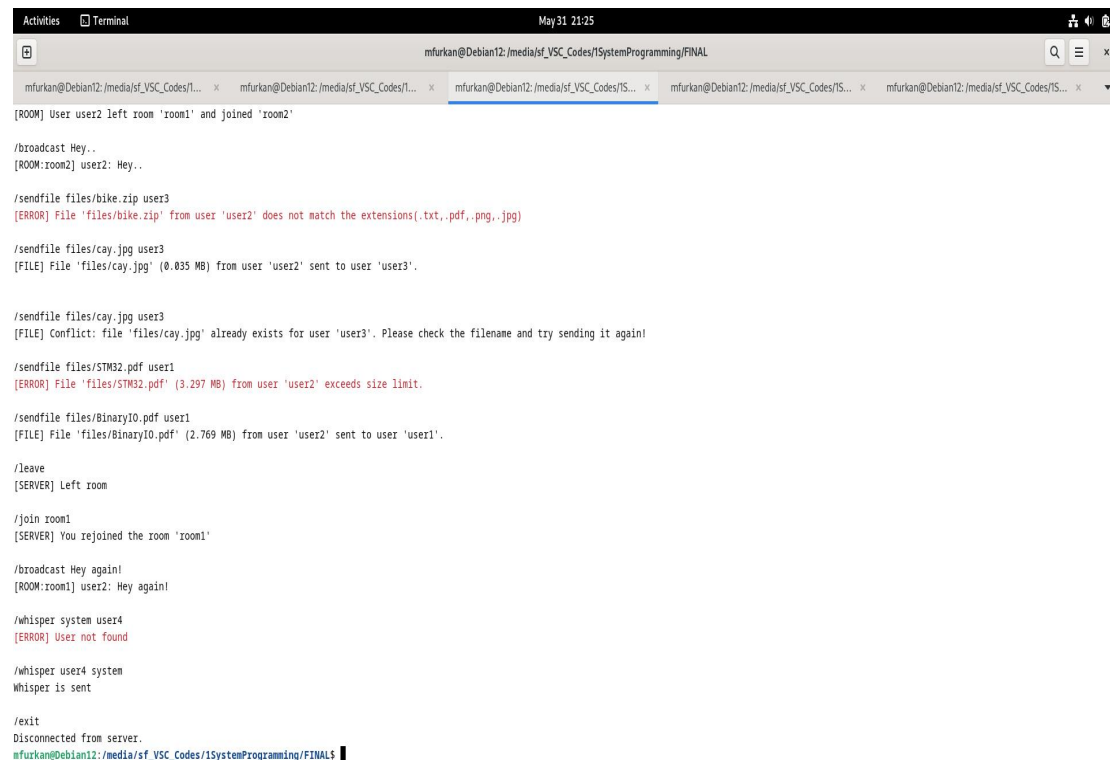
/sendfile files/cay.jpg user3
[FILE] Conflict: file 'files/cay.jpg' already exists for user 'user3'. Please check the filename and try sending it again!

/sendfile files/STM32.pdf user1
[ERROR] File 'files/STM32.pdf' (3.297 MB) from user 'user2' exceeds size limit.

/sendfile files/BinaryIO.pdf user1
[FILE] File 'files/BinaryIO.pdf' (2.769 MB) from user 'user2' sent to user 'user1'.

/leave
[SERVER] Left room

/join room1
[SERVER] You rejoined the room 'room1'
```



A terminal window titled 'May 31 21:25' showing the output of a chat client. The user is in a room named 'room1' and has sent several messages. The chat client is connected to a server at 127.0.0.1:4000. The output shows the user's messages and the server's responses, including file upload errors and room management actions.

```
mfurkan@Debian12:/media/sf_VSC_Codes/ISystemProgramming/FINAL$ ./chatclient 127.0.0.1 4000

[ROOM] User user2 left room 'room1' and joined 'room2'

/broadcast Hey..
[ROOM:room2] user2: Hey..

/sendfile files/bike.zip user3
[ERROR] File 'files/bike.zip' from user 'user2' does not match the extensions(.txt,.pdf,.png,.jpg)

/sendfile files/cay.jpg user3
[FILE] File 'files/cay.jpg' (0.035 MB) from user 'user2' sent to user 'user3'.

/sendfile files/cay.jpg user3
[FILE] Conflict: file 'files/cay.jpg' already exists for user 'user3'. Please check the filename and try sending it again!

/sendfile files/STM32.pdf user1
[ERROR] File 'files/STM32.pdf' (3.297 MB) from user 'user2' exceeds size limit.

/sendfile files/BinaryIO.pdf user1
[FILE] File 'files/BinaryIO.pdf' (2.769 MB) from user 'user2' sent to user 'user1'.

/leave
[SERVER] Left room

/join room1
[SERVER] You rejoined the room 'room1'

/broadcast Hey again!
[ROOM:room1] user2: Hey again!

/whisper system user4
[ERROR] User not found

/whisper user4 system
Whisper is sent

/exit
Disconnected from server.
mfurkan@Debian12:/media/sf_VSC_Codes/ISystemProgramming/FINAL$
```

```
Activities Terminal May 31 21:25
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL$ ./chatclient 127.0.0.1 4000
[INFO] Connected to server.
Enter username (max 16 chars, alphanumeric): user1
/join room1
[SERVER] You joined the room 'room1'

/broadcast Hello!
[ROOM:room1] user1: Hello!

/whisper user2 psst
Whisper is sent

[FILE] File 'files/Binary10.pdf' (2.769 MB) from user 'user2' sent to user 'user1'.

[ROOM:room1] user2: Hey again!

/exit
Disconnected from server.
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL$
```

```
Activities Terminal May 31 21:26
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL$ make
gcc -Wall -pthread -o chatclient client/chatclient.c
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL$ ./chatserver 4000
[START] Server listening on port 4000...
[CONNECT] New Client connected: 'user1' connected from 127.0.0.1
[CONNECT] New Client connected: 'user2' connected from 127.0.0.1
[CONNECT] New Client connected: 'user3' connected from 127.0.0.1
[CONNECT] New Client connected: 'user4' connected from 127.0.0.1
[COMMAND] user1 joined room 'room1'
[COMMAND] user2 joined room 'room1'
[COMMAND] user3 joined room 'room2'
[COMMAND] user4 joined room 'room1'
[COMMAND] user 'user1' broadcasted to room1
[COMMAND] user1 whispered -> user2: psst
[COMMAND] user2 joined room 'room2' from room 'room1'
[COMMAND] user 'user2' broadcasted to room2
[ERROR] File 'files/bike.zip' from user 'user2' does not match the extensions(.txt,.pdf,.png,.jpg)
[FILE] Conflict: file 'files/cay.jpg' already exists for user 'user3'. Please check the filename and try sending it again!

[ERROR] File 'files/STM32.pdf' (3.297 MB) from user 'user2' exceeds size limit.
[COMMAND] user 'user2' left room 'room2'
[COMMAND] user2 rejoined room 'room1'
[COMMAND] user 'user2' broadcasted to room1
[ERROR] User 'system' not found
[COMMAND] user2 whispered -> user4: system
[DISCONNECT] user 'user2' disconnected
[DISCONNECT] user 'user1' disconnected
^C[SHUTDOWN] Signal received! Server is shut down! Disconnecting all 2 clients...
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL$
```

```
Activities Terminal May 31 21:26
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL
mfurkan@Debian12: /media/sf_VSC_Codes/1... mfurkan@Debian12: /media/sf_VSC_Codes/1... mfurkan@Debian12: /media/sf_VSC_Codes/1S... mfurkan@Debian12: /media/sf_VSC_Codes/1S... mfurkan@Debian12: /media/sf_VSC_Codes/1S...
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL$ ./chatclient 127.0.0.1 4000

[INFO] Connected to server.
Enter username (max 16 chars, alphanumeric): user3
/join room2
[SERVER] You joined the room 'room2'

[ROOM:room2] user2: Hey..

[FILE] File 'files/cay.jpg' (0.035 MB) from user 'user2' sent to user 'user3'.

[FILE] Conflict: file 'files/cay.jpg' already exists for user 'user3'. Please check the filename and try sending it again!

[INFO] Server shutting down

[INFO] Server has closed the connection.
Please press Enter to exit.

Disconnected from server.
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL$
```

```
Activities Terminal May 31 21:26
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL
mfurkan@Debian12: /media/sf_VSC_Codes/1... mfurkan@Debian12: /media/sf_VSC_Codes/1... mfurkan@Debian12: /media/sf_VSC_Codes/1S... mfurkan@Debian12: /media/sf_VSC_Codes/1S... mfurkan@Debian12: /media/sf_VSC_Codes/1S...
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL$ ./chatclient 127.0.0.1 4000

[INFO] Connected to server.
Enter username (max 16 chars, alphanumeric): user4
/join room1
[SERVER] You joined the room 'room1'

[ROOM:room1] user1: Hello!

[ROOM:room1] user2: Hey again!

[WHISPER] user2: system

[INFO] Server shutting down

[INFO] Server has closed the connection.
Please press Enter to exit.

Disconnected from server.
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL$
```

```
2025-05-31 21:20:37 - [START] Server listening on port 4000...
2025-05-31 21:21:05 - [CONNECT] New Client connected: 'user1' connected from 127.0.0.1
2025-05-31 21:21:08 - [CONNECT] New Client connected: 'user2' connected from 127.0.0.1
2025-05-31 21:21:11 - [CONNECT] New Client connected: 'user3' connected from 127.0.0.1
2025-05-31 21:21:13 - [CONNECT] New Client connected: 'user4' connected from 127.0.0.1
2025-05-31 21:21:18 - [JOIN] user 'user1' joined room 'room1'
2025-05-31 21:21:22 - [JOIN] user 'user2' joined room 'room1'
2025-05-31 21:21:27 - [JOIN] user 'user3' joined room 'room2'
2025-05-31 21:21:32 - [JOIN] user 'user4' joined room 'room1'
2025-05-31 21:21:40 - [BROADCAST] user 'user1' broadcasted Hello!
2025-05-31 21:21:52 - [WHISPER] user1 whispered -> user2: psst
2025-05-31 21:22:06 - [JOIN] user 'user2' joined room 'room2' from room 'room1'
2025-05-31 21:22:17 - [BROADCAST] user 'user2' broadcasted Hey..
2025-05-31 21:22:34 - [ERROR] File 'files/bike.zip' from user 'user2' does not match the extensions(.txt,.pdf,.png,.jpg)
2025-05-31 21:22:48 - [FILE] File 'files/cay.jpg' (0.035 MB) from user 'user2' sent to user 'user3'.
2025-05-31 21:23:25 - [FILE] Conflict: file 'files/cay.jpg' already exists for user 'user3'. Please check the filename and try sending it again!
2025-05-31 21:23:41 - [ERROR] File 'files/STM32.pdf' (3.297 MB) from user 'user2' exceeds size limit.
2025-05-31 21:23:57 - [FILE] File 'files/BinaryIO.pdf' (2.769 MB) from user 'user2' sent to user 'user1'.
2025-05-31 21:24:18 - [LEAVE] user 'user2' left room 'room2'
2025-05-31 21:24:23 - [JOIN] user 'user2' rejoined room 'room1'
2025-05-31 21:24:31 - [BROADCAST] user 'user2' broadcasted Hey again!
2025-05-31 21:24:48 - [ERROR] User 'system' not found
2025-05-31 21:24:58 - [WHISPER] user2 whispered -> user4: system
2025-05-31 21:25:09 - [DISCONNECT] user 'user2' disconnected
2025-05-31 21:25:52 - [DISCONNECT] user 'user1' disconnected
2025-05-31 21:25:59 - [SHUTDOWN] SIGINT received. Disconnecting clients.
```

```
Enter username (max 16 chars, alphanumeric): user1
/join room1
[SERVER] You joined the room 'room1'

/broadcast hey
[ROOM:room1] user1: hey

[ROOM:room1] user3: system

[WHISPER] user2: psst

/sendfile files/dnm.txt user2
[ERROR] File 'files/dnm.txt' does not exist

/sendfile files/baklv.jpg user2
[FILE] File 'files/baklv.jpg' (0.368 MB) from user 'user1' sent to user 'user2'.

/sendfile files/baklv.jpg user2
[FILE] Conflict: file 'files/baklv.jpg' already exists for user 'user2'. Please check the filename and try sending it again!

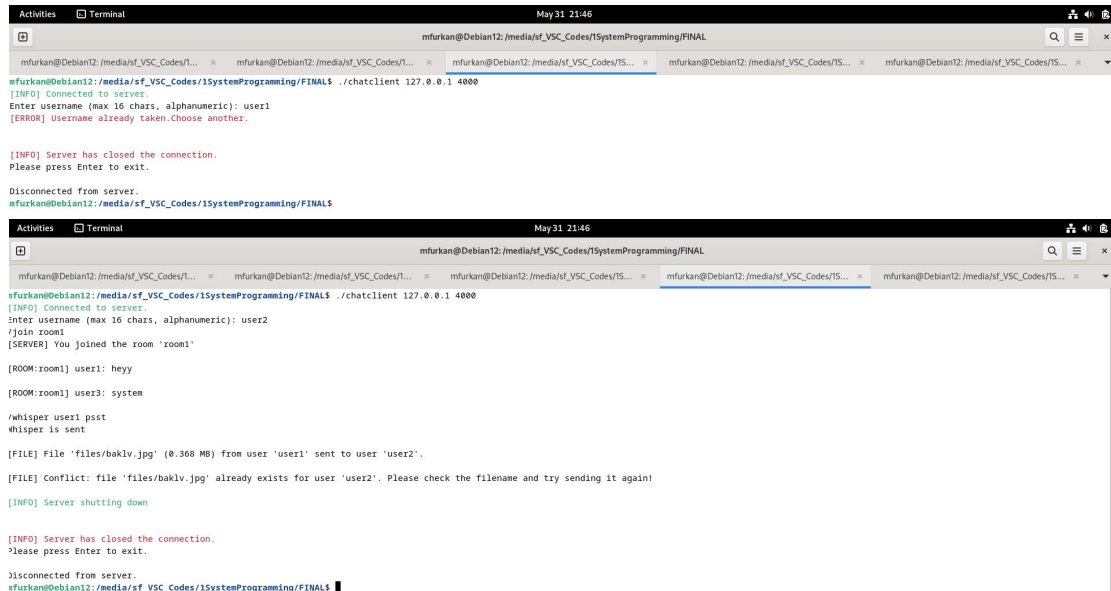
/sendfile files/baklv.jpg user3
[FILE] File 'files/baklv.jpg' (0.368 MB) from user 'user1' sent to user 'user3'.

[FILE] File 'files/redc.png' (0.522 MB) from user 'user3' sent to user 'user1'.

/whisper
[ERROR] Unknown command

/whisper fg
[ERROR] Usage: /whisper <username> <message>
/sdf
[ERROR] Unknown command

/exit
Disconnected from server.
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL$
```



```
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL$ ./chatclient 127.0.0.1 4000
[INFO] Connected to server.
Enter username (max 16 chars, alphanumeric): user1
[ERROR] username already taken.Choose another.

[INFO] Server has closed the connection.
Please press Enter to exit.

Disconnected from server.
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL$
```

```
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL$ ./chatclient 127.0.0.1 4000
[INFO] Connected to server.
Enter username (max 16 chars, alphanumeric): user2
/join room1
[SERVER] You joined the room 'room1'

[ROOM:room1] user1: hey

[ROOM:room1] user3: system

/whisper user1 psst
whisper is sent

[FILE] File 'files/baklv.jpg' (0.368 MB) from user 'user1' sent to user 'user2'.

[FILE] Conflict: file 'files/baklv.jpg' already exists for user 'user2'. Please check the filename and try sending it again!

[INFO] Server shutting down

[INFO] Server has closed the connection.
Please press Enter to exit.

Disconnected from server.
mfurkan@Debian12: /media/sf_VSC_Codes/1SystemProgramming/FINAL$
```

```
Activities Terminal May 31 21:46
mfurkan@Debian12:/media/sf_VSC_Codes/1SystemProgramming/FINAL$
mfurkan@Debian12:/media/sf_VSC_Codes/1... x mfurkan@Debian12:/media/sf_VSC_Codes/1... x mfurkan@Debian12:/media/sf_VSC_Codes/1S... x mfurkan@Debian12:/media/sf_VSC_Codes/1S... x mfurkan@Debian12:/media/sf_VSC_Codes/1S... x
mfurkan@Debian12:/media/sf_VSC_Codes/1SystemProgramming/FINAL$ ./chatclient 127.0.0.1 4000
[INFO] Connected to server.
Enter username (max 16 chars, alphanumeric): user3
/join room2
[SERVER] You joined the room 'room2'

/join room1
[ROOM] User user3 left room 'room2' and joined 'room1'

/broadcast system
[ROOM:room1] user3: system

[FILE] File 'files/baklv.jpg' (0.368 MB) from user 'user1' sent to user 'user3'.

/sendfile files/redc.png user1
[FILE] File 'files/redc.png' (0.522 MB) from user 'user3' sent to user 'user1'.

[INFO] Server shutting down

[INFO] Server has closed the connection.
>please press Enter to exit.

$!
disconnected from server.
mfurkan@Debian12:/media/sf_VSC_Codes/1SystemProgramming/FINAL$
```

```
Activities Terminal May 31 21:46
mfurkan@Debian12:/media/sf_VSC_Codes/1SystemProgramming/FINAL$
mfurkan@Debian12:/media/sf_VSC_Codes/1... x mfurkan@Debian12:/media/sf_VSC_Codes/1... x mfurkan@Debian12:/media/sf_VSC_Codes/1S... x mfurkan@Debian12:/media/sf_VSC_Codes/1S... x mfurkan@Debian12:/media/sf_VSC_Codes/1S... x
mfurkan@Debian12:/media/sf_VSC_Codes/1SystemProgramming/FINAL$ ./chatserver 4000
[START] Server listening on port 4000...
[CONNECT] New Client connected: 'user1' connected from 127.0.0.1
[CONNECT] New Client connected: 'user2' connected from 127.0.0.1
[CONNECT] New Client connected: 'user3' connected from 127.0.0.1
[COMMAND] user3 joined room 'room2'
[COMMAND] user2 joined room 'room1'
[COMMAND] user1 joined room 'room1'
[COMMAND] user 'user1' broadcasted to room1
[COMMAND] user3 joined room 'room1' from room 'room2'
[COMMAND] user 'user3' broadcasted to room1
[COMMAND] user2 whispered -> user1: psst
[ERROR] File 'files/dnm.txt' does not exist
[FILE] Conflict: file 'files/baklv.jpg' already exists for user 'user2'. Please check the filename and try sending it again!

[DISCONNECT] user 'user1' disconnected
^C[SHUTDOWN] Signal received! Server is shut down! Disconnecting all 2 clients...
mfurkan@Debian12:/media/sf_VSC_Codes/1SystemProgramming/FINAL$
```

```
2025-05-31 21:41:52 - [START] Server listening on port 4000...
2025-05-31 21:41:58 - [CONNECT] New Client connected: 'user1' connected from 127.0.0.1
2025-05-31 21:42:03 - [REJECTED] Duplicate username attempted: 'user1'

2025-05-31 21:42:09 - [CONNECT] New Client connected: 'user2' connected from 127.0.0.1
2025-05-31 21:42:19 - [CONNECT] New Client connected: 'user3' connected from 127.0.0.1
2025-05-31 21:42:24 - [JOIN] user 'user3' joined room 'room2'
2025-05-31 21:42:29 - [JOIN] user 'user2' joined room 'room1'
2025-05-31 21:42:34 - [JOIN] user 'user1' joined room 'room1'
2025-05-31 21:42:40 - [BROADCAST] user 'user1' broadcasted hey!
2025-05-31 21:42:55 - [JOIN] user 'user3' joined room 'room1' from room 'room2'
2025-05-31 21:43:06 - [BROADCAST] user 'user3' broadcasted system
2025-05-31 21:43:18 - [WHISPER] user2 whispered -> user1: psst
2025-05-31 21:43:39 - [ERROR] File 'files/dnm.txt' does not exist

2025-05-31 21:43:51 - [FILE] File 'files/baklv.jpg' (0.368 MB) from user 'user1' sent to user 'user2'.

2025-05-31 21:44:01 - [FILE] Conflict: file 'files/baklv.jpg' already exists for user 'user2'. Please check the filename and try sending it again!

2025-05-31 21:44:11 - [FILE] File 'files/baklv.jpg' (0.368 MB) from user 'user1' sent to user 'user3'.

2025-05-31 21:44:31 - [FILE] File 'files/redc.png' (0.522 MB) from user 'user3' sent to user 'user1'.

2025-05-31 21:45:36 - [DISCONNECT] user 'user1' disconnected
2025-05-31 21:45:38 - [SHUTDOWN] SIGINT received. Disconnecting clients.
```

4. Conclusion & Potential Improvements

Conclusion

The system successfully implements:

- Multi-room chat with whisper support.
- Thread-safe file transfers.
- Graceful disconnection handling.

Improvements

Area	Suggested Improvement
● Scalability	Replace thread-per-client with an event loop (epoll/kqueue)
● Security	Add TLS encryption for secure messaging
● File Transfers	Implement resumable transfers
● Logging	Use syslog for better log management
● Mobile Support	Add a REST API for mobile clients

Final Thoughts

The project demonstrates robust multi-threaded socket programming with proper error handling and resource management. Future work could focus on scalability and security enhancements.

Screenshots